

UNICAMP

UNIVERSIDADE ESTADUAL DE CAMPINAS (UNICAMP)

INSTITUTO DE MATEMÁTICA, ESTATÍSTICA E COMPUTAÇÃO CIENTÍFICA (IMECC)

PROGRAMA DE PÓS-GRADUAÇÃO EM ESTATÍSTICA

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Title of the dissertation/thesis

Título da dissertação/tese (em português)

Campinas

Ano

Caio Gomes Alves

Title of the dissertation/thesis

Título da dissertação/tese (em português)

Dissertação apresentada ao Instituto de Matemática, Estatística e Computação Científica da Universidade Estadual de Campinas (UNICAMP) como parte dos requisitos exigidos para a obtenção do título de Mestre em Estatística.

Dissertation presented to the Institute of Mathematics, Statistics and Scientific Computing of the University of Campinas (UNICAMP) as part of the requirements for the degree of Master in Statistics.

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Este trabalho corresponde à versão final da Dissertação defendida.

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A folha de aprovação será fornecida pela Secretaria de Pós-Graduação

Este trabalho é dedicado às crianças adultas que,
quando pequenas, sonharam em se tornar cientistas.

Acknowledgements

Inserir aqui os agradecimentos. ATENÇÃO ALUNO BOLSISTA: você precisa mencionar agradecimento ao órgão de fomento de sua bolsa, incluindo explicitamente o código do processo. Caso seja bolsista CAPES, especial atenção ao Art. 3º da Portaria CAPES 206/2018, que prevê texto padrão de agradecimento.

"If you can solve your problem, then what is the need of worrying?
If you cannot solve it, then what is the use of worrying"
Shantideva

Resumo

Segundo a ABNT (2003, 3.1-3.2), o resumo deve ressaltar o objetivo, o método, os resultados e as conclusões do documento. A ordem e a extensão destes itens dependem do tipo de resumo (informativo ou indicativo) e do tratamento que cada item recebe no documento original. O resumo deve ser precedido da referência do documento, com exceção do resumo inserido no próprio documento. Neste trabalho, devem ser utilizadas até 500 palavras. (. . .) As palavras-chave devem figurar logo abaixo do resumo, antecedidas da expressão Palavras-chave:, separadas entre si por ponto e finalizadas também por ponto.

Palavras-chave: Latex. Abntex. Editoração de texto.

Abstract

This is the english abstract.

Keywords: Latex. Abntex. Text editoration.

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List of abbreviations and acronyms

UNICAMP	Universidade Estadual de Campinas
IMECC	Instituto de Matemática, Estatística e Computação Científica
LabCSD	Laboratório de Controle e Sistemas Dinâmicos
EPIFISMA	Laboratório de Epidemiologia e Fisiologia Matemática
LCP	Laboratório de Computação Paralela
LGC	Laboratório de Geofísica Computacional
LMDC	Laboratório de Matemática Discreta e Códigos
MiLAB	Laboratório de Tratamento Matemático de Imagens e Inteligência Computacional
LPOO	Laboratório de Pesquisa Operacional e Otimização
LEM	Laboratório de Ensino de Matemática
PAPMEM	Programa de Aperfeiçoamento para Professores de Matemática do Ensino Médio
OMU	Olimpíada de Matemática da Unicamp
CCPG	Comissão Central de Pós-Graduação
ABNT	Associação Brasileira de Normas Técnicas
abnTeX	ABsurdas Normas para TeX

List of symbols

$\mathbb{M}_{m\times n}(\mathbb{R})$	Conjunto das matrizes de ordem $m \times n$ com entradas reais
$\mathbb{R}^n_+$	Conjunto dos vetores x pertencentes a $\mathbb{R}^n$ que satisfazem $x \geqslant 0$
$\mathbb{R}^n_{++}$	Conjunto dos vetores x pertencentes a $\mathbb{R}^n$ que satisfazem $x > 0$
$\|\cdot\|$	Norma-p vetorial
$\|\|\|\cdot\|\|\|_p$	Norma-p matricial
$\mathbf{I}_n$	Matriz identidade de ordem $n \times n$
∇f	Gradiente da função f
∞	Infinito

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Introduction

Este documento e seu código-fonte são exemplos de referência de uso da classe abntex2 e do pacote abntex2cite. O documento exemplifica a elaboração de trabalho acadêmico (teses e dissertações) produzido inicialmente conforme a **Informação CCPG/001/2015** (que trata das *Normas para impressão de teses/dissertações* da UNICAMP). Posteriormente, foi adequada considerando a Instrução Normativa CCPG/06/2025 (CCPG–UNICAMP, 2025). Encorajamos o leitor a consultar estas instruções antes de iniciar as alterações neste documento e seu código-fonte.

A elaboração deste modelo teve como base uma customização do "Modelo Canônico de Trabalho Acadêmico com abnTEX2" (ARAUJO, 2015b) para que as normas presentes na Informação CCPG/001/2015 fossem respeitadas. O modelo original produzido pela equipe abnTEX2 cumpre as seguintes normas ABNT:

1. **ABNT NBR 14724:2011**: Informação e documentação - Trabalhos acadêmicos - Apresentação;
2. **ABNT NBR 10520:2002**: Informação e documentação - Citações;
3. **ABNT NBR 6034:2004**: Informação e documentação - Índice - Apresentação;
4. **ABNT NBR 6028:2003**: Informação e documentação - Resumo - Apresentação;
5. **ABNT NBR 6027:2012**: Informação e documentação - Sumário - Apresentação;
6. **ABNT NBR 6024:2012**: Informação e documentação - Numeração progressiva das seções de um documento - Apresentação
7. **ABNT NBR 6023:2002**: Informação e documentação - Referência - Elaboração.

Este documento deve ser utilizado como complemento dos manuais do abnTEX2 (ARAUJO, 2015a; ARAUJO, 2015c; ARAUJO, 2015d) e da classe memoir (WILSON; MADSEN, 2010).

A leitura do teor desde documento (tanto o PDF quando os arquivos que compõem seu código-fonte), bem como do arquivo LEIAME.txt é altamente recomendada para melhor entendimento da dinâmica de funcionamento da classe abntex2 e do pacote abntex2cite. Seus principais comandos e usos estão exemplificados no decorrer do texto, bem como outras informações relevantes para a escrita de seu trabalho acadêmico.

Part I

Theoretical foundations

1 Spatio-Temporal Point Processes

1.1 Marked Spatio-Temporal Point Processes

A spatio-temporal point process $\mathcal{X}$ is a sequence of random variables on $\{S \subset \mathbb{R}^2 \times [0,T]\}$ in which the number of points in $S \times T$ has a counting measure $N(S \times T)$ which is almost surely finite, when $S \times T$ is bounded. Usually $\mathcal{X}$ is observed in a spatio-temporal window $W = S \times T$, and a realization of $\mathcal{X}$ in W is a set $\mathbf{x} = \{(s_i, t_i), i = 1, 2, \ldots, n\}$, where $s_i = (x_i, y_i)$ and t_i represent the spatial coordinates and times of occurrence of the i-th event, respectively.

A marked spatio-temporal point process incorporates a uniquely associated component m_i with each event of $\mathcal{X}$. In earthquake's catalogs, the marks m_i usually denote the magnitude of each event, while (s_i, t_i) represents the longitude/latitude of the earthquake's epicenter and the ordered times of occurrence.

Table 1 – Example of an earthquake catalog.

Event	**Time**	**Longitude**	**Latitude**	**Magnitude**
1	t_1	x_1	y_1	m_1
2	t_2	x_2	y_2	m_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
n	t_n	x_n	y_n	m_n

Consider the event history $\mathcal{H}_t = \{(x_i, y_i, t_i, m_i) \in \mathcal{X} : t_i < t\}$. A spatio-temporal point process can be fully characterized by it's conditional intensity function $\lambda(s, t|\mathcal{H}_t)$ (DALEY; VERE-JONES, 2003)

$$\lambda(s, t|\mathcal{H}_t) = \lim_{\delta s, \delta t \to 0} \frac{\mathbb{E}\left[N(\delta s \times \delta t)|\mathcal{H}_t\right]}{\delta s \, \delta t}, \tag{1.1}$$

that represents the expected number of events observed in an infinitesimal region of volume $(\delta s \times \delta t)$ centered around (s, t), conditional on the observed history of the process up to time t. With it, we can define the process's associated compensator $\Lambda(t)$

$$\Lambda(t|\mathcal{H}_u) = \int_0^t \int_S \lambda(s, u|\mathcal{H}_u) ds \, du, \tag{1.2}$$

in which $N(s, t|\mathcal{H}_t) - \Lambda(t|\mathcal{H}_t)$ is an local $\mathcal{H}_t$-martingale. The simplest and most common kind of spatio-temporal point process is the Poisson Process, defined as such:

Definition 1 (Poisson Process)**.** *A spatio-temporal counting process* $\{N(s,t) : s \in \mathbb{R}^2, t \in \mathbb{R}_+\}$ *is a Inhomogeneous Poisson Process with intensity function* $\lambda(s,t)$ *if:*

- *For any bounded window* $W = (S \times T)$, $N(W) \sim \text{Pois}\left(\int_T \int_S \lambda(s,t) ds\, dt\right)$;
- *For any* n *disjoint windows* $W_1, W_2, \ldots, W_n$, *the random variables* $N(W_1), N(W_2), \ldots, N(W_n)$ *are mutually independent.*

The Poisson Process has two properties that makes it very versatile for modeling point processes in general: **Superposition** and **Thinning**:

Theorem 1 (Superposition of Poisson Processes)**.** *Given* k *independent Poisson Processes* $\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_k$ *with intensity functions* $\lambda_1, \lambda_2, \ldots, \lambda_k$, *the new process* $\mathcal{X} = \sum_{i=1}^{k} \mathcal{X}_i$ *is a Poisson Process, with intensity function* $\lambda = \sum_{i=1}^{k} \lambda_i$.

Theorem 2 (Thinning of Poisson Processes)**.** *Let* $\mathcal{X}$ *be a Poisson Process with intensity function* λ. *For each occurrence of* $\mathcal{X}$, *assign it to process* $\mathcal{X}_1$ *with probability* p_1, *to process* $\mathcal{X}_2$ *with probability* p_2, *and so on up to process* $\mathcal{X}_k$ *with probability* p_k, *such that* $\sum_{i=1}^{k} p_i = 1$. *The new processes* $\{\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_k\}$ *are Poisson Processes with intensity functions* $\{p_1\lambda, p_2\lambda, \ldots, p_k\lambda\}$, *and are mutually independent.*

With these two results, many point processes can have their intensity function modelled using Poisson Processes. However, one condition necessary for that is that $\lambda(s,t)$ must remain constant along time (i.e. $\lambda(s,t) = \lambda(s)$, $\forall t \in [0,T]$), and in many cases that is not necessarily true.

1.2 Hawkes Processes and the ETAS Model

In many point phenomena, the ocurrence of events increases the probability of further events occuring, such as the spread of diseases, urban violence (Mohler et al. (2011), Ludwig et al. (2025)) and earthquakes. For these kinds of processes, the intensity function must change dynamically, in order to accommodate the increase in probability triggered by a previous event.

A self-exciting point process (occuring in time) was proposed by Hawkes (1971a), having intensity function given by

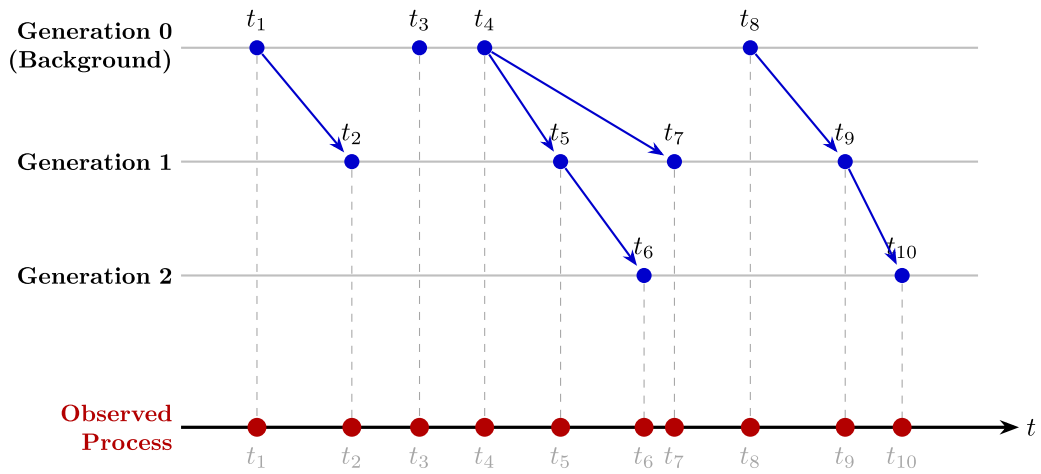
$$\lambda(t|\mathcal{H}_t) = \mu + \sum_{i:t_i<t} g(t - t_i), \tag{1.3}$$

where $\mu > 0$ is the baseline intensity of the process, and $\sum_{i:t_i<t} g(t-t_i)$ represents the contribution of past events to the increase in the conditional intensity up to time t, with

$g(v) \geqslant 0$ for nonnegative v and 0 otherwise. To guarantee stationarity for this process (Hawkes (1971b)), is necessary that $\int_0^\infty g(v)dv < 1$, otherwise the process can have infinite events in finite time.

The **ETAS** (*Epidemic-Type Aftershock Sequence*) Model was proposed by Ogata (1988), and is the most used to describe earthquake occurrences. The initial model was only concerned with the event times and was a particular case of a marked point process, for it's inclusion of the earthquake's magnitudes. This model assumes that any earthquake can come from one of two sources: the **background** events come from a Inhomogeneous Poisson Process, independent of time, and the **offspring** were triggered by a previous event (either a background, or another offspring). Another way to call these events are **aftershocks**, and have a more geological interpretation. Figure 1 shows the branching structure of a temporal **ETAS**, with events being parents if they come from generation 0, and offsprings if they come from any other generation.

Figure 1 – Branching of a self-exciting point process.



In a later work, Ogata (1998) extended the ETAS model to accommodate the spatio-temporal distribution of earthquakes, in the form

$$\lambda_\theta(x, y, t|\mathcal{H}_t) = \mu(x, y) + \sum_{i:t_i<t} \kappa(m_i)g(t - t_i; m_i)f(x - x_i, y - y_i; m_i), \tag{1.4}$$

with it's components being:

- $\mu(x, y)$ is the background rate, representing a smooth function on the region S, with $\mu(x, y) > 0$. The parent events occur following this background rate, and it is assumed to be independent of time.

- $\kappa(m_i)$ is the productivity of the process, representing the expected number of events triggered from a event of magnitude m_i, taking form

$$\kappa(m) = A \exp[\alpha(m - m_0)],$$

with $A > 0$ and $\alpha > 0$ being unknown parameters, and m_0 the cutoff magnitude being considered in the study (i.e., all the n events in the catalog have magnitudes $m_i \geqslant m_0, i = 1, 2, \ldots, n$).

- $g(t - t_i)$ is the probability density function of the time (in days) of a triggered event occuring at time t_i, and is assumed to be independent of the event magnitude m_i. A power law density is considered, based in the modified Omori's law (Utsu et al. (1995))

$$g(t - t_i) = \begin{cases} (p-1)c^{p-1}\left((t - t_i) + c\right)^{-p}, & t - t_i > 0 \\ 0, & t - t_i \leqslant 0 \end{cases},$$

with $c > 0$ and $p > 1$ being unknown parameters.

- $f(x - x_i, y - y_i; m_i)$ is the probability density function of the location in space of a event triggered from a event occurred in (x_i, y_i) and magnitude m_i. The distances are considered to be in degrees[2] (x is the longitude and y is the latitude), and this can cause some considerable distortion in areas with big latitudes. An isotropic density function is considered

$$f(x - x_i, y - y_i; m_i) = \frac{q-1}{\pi D \exp[\gamma(m_i - m_0)]} \left(1 + \frac{(x - x_i)^2 + (y - y_i)^2}{D \exp[\gamma(m_i - m_0)]}\right)^{-q},$$

with $D > 0, \gamma > 0$ and $q > 1$ are unknown parameters.

Other parametrizations for $\lambda_\theta(x, y, t|\mathcal{H}_t)$ were proposed by Ogata, but this one offers great flexibility by allowing the times of triggered events to be independent of the magnitudes, and by having separated parameters that control different aspects of the earthquake's intensity function behavior (temporal decay rate, influence of magnitudes in the number of expected offspring, spatial clustering, etc). The model can also incorporate the distribution of the marks/magnitudes

$$\lambda_{\theta,\beta}(x, y, t, m|\mathcal{H}_t) = \nu(m)\lambda_\theta(x, y, t|\mathcal{H}_t), \tag{1.5}$$

where $\nu(m)$ follows the Gutenberg-Richter distribution (Gutenberg & Richter (1944))

$$\nu(m) = \beta \exp[-\beta(m - m_0)], \ \beta > 0.$$

The expected number of offsprings generated from any event, regardless of its magnitude is given by

$$e_{\text{off}} = \int_{m_0}^{\infty} \kappa(m)\nu(m)dm = \frac{A\beta}{\beta - \alpha}. \tag{1.6}$$

If $e_{\text{off}} < 1$, the model is stable (Daley & Vere-Jones (2003)), and has cluster size of $\frac{1}{(1-e_{\text{off}})} = \frac{\beta-\alpha}{\beta(1-A)-\alpha}$. Because the marks distribution is independent of the other parameters (as the complete intensity $\lambda_{\theta,\beta}(x,y,t,m|\mathcal{H}_t)$ is the product of $\nu(v)$ and $\lambda_\theta(x,y,t|\mathcal{H}_t)$), the estimation of β can be done without the need to estimate the other parameters.

1.3 Likelihood and Stochastic Declustering

Estimating the model parameters β and $\theta = (A, \alpha, c, p, D, \gamma, q)$ and the background intensity function $\mu(x,y)$ is crucial to understand the seismological dynamic of a certain region, as it dictates when and where the aftershocks of an earthquake have more probability of happening. Moreover, early-warning systems depend on this estimates to make predictions in feasible time, as the time needed for these systems to mitigate further damages are between 1 and 10 minutes (Allen & Melgar (2019)).

For this, maximum likelihood methods for estimating the vector parameters have many desirable properties, such as being consistent, asymptotically unbiased and efficient (Ogata (1978)). The ETAS model complete log-likelihood is given by

$$l(\theta,\beta|\mathcal{H}_t) = \sum_{i=1}^{n} \log(\lambda_{\theta,\beta}(x_i,y_i,t_i,m_i|\mathcal{H}_t)) - \int_{m_0}^{\infty}\int_0^T\int\int_S \lambda_{\theta,\beta}(x,y,t,m|\mathcal{H}_t)dx\,dy\,dt\,dm, \tag{1.7}$$

where T is the time of the last event in the catalog (the time integral starts at 0 because we can re-scale the event times $(t_1,t_2,\ldots,t_n)$ such that we have the fist event occur at time 0 with $(t_1-t_1, t_2-t_1,\ldots,t_n-t_1) = (0,t_2^*,\ldots,T)$). As seen in Equation; (1.5), β and θ are separable, and $\int_{m_0}^{\infty}\nu(m)dm = 1$ (because it is a probability density function), we can decompose the previous log-likelihood as

$$l(\theta,\beta|\mathcal{H}_t) = l_1(\beta|\mathcal{H}_t) + l_2(\theta|\mathcal{H}_t)$$

where

$$l_1(\beta|\mathcal{H}_t) = \sum_{i=1}^{n}\log(\nu(m_i)) = n\log(\beta) - \beta\sum_{i=1}^{n}(m_i - m_0) \tag{1.8}$$

and

$$l_2(\theta|\mathcal{H}_t) = \sum_{i=1}^{n}\log(\lambda_\theta(x_i,y_i,t_i|\mathcal{H}_t)) - \int_0^T\int\int_S \lambda_\theta(x,y,t|\mathcal{H}_t)dx\,dy\,dt. \tag{1.9}$$

The maximum likelihood estimator of β can be obtained analytically using Equation (1.8)

$$\hat{\beta} = \frac{n}{\sum_{i=1}^{n}(m_i - m_0)},$$

whereas the maximization of Equation (1.9) requires some form of numerical optimization method, to approximate the integral part. When we evaluate the integral term, we have

$$\int_0^T \int \int_S \lambda_\theta(x, y, t|\mathcal{H}_t) = T \int \int_S \mu(x, y)dx\, dy \; + \sum_{i:t_i<t} \kappa(m_i)G(T - t_i)F(S - s_i; m_i)$$

where $G(t)$ and $F(s)$ are the cumulative probability functions of $g(t)$ and $f(s)$, respectively, and are defined in Appendix A. For any event not occurring near the spatio-temporal border (so that all the probability of being a offspring comes from the events inside the observed window, and not from any external event), we have that $G(T - t_i) \approx 1$ and $F(S - s_i) \approx 1$. This approximation simplifies the integral, making the log-likelihood easier to evaluate, when the assumptions are met.

However, the direct optimization of the complete log-likelihood remains complicated, due to two main factors: the $\log(\lambda_\theta(x_i, y_i, t_i|\mathcal{H}_t))$ term is the logarithm of a sum that has all the parameters and is not separable, and the computational cost for each evaluation is in the order of $\mathcal{O}(n^2)$. If the underlying branching structure were known, the log-likelihood would be separable because the terms for the contribution of the background events and the offsprings would be either 1 or 0 for any occurrence.

Consider that each event j has a parent $u_j \in \{0, 1, \ldots, j-1\}$, in which $u_j = 0$ indicates a event that has come from the background rate function. If u_j were observed, we would have

$$l_2(\theta|\mathcal{H}_t) = \sum_{j=1}^{n} \left[\mathbb{1}(u_j = 0) \log(\mu(x_j, y_j)) + \sum_{i<j} \mathbb{1}(u_j \neq 0) \log(\kappa(m_i)g(t_j - t_i)f(s_j - s_i; m_i)) \right] - T \int \int_S \mu(x, y)dx\, dy + \sum_{i:t_i<t} \kappa(m_i)G(T - t_i)F(S - s_i; m_i), \tag{1.10}$$

making the log-likelihood separable, and with direct evaluation of the parameters. Zhuang et al. (2002) proposed a stochastic declustering method based on the decomposition of the conditional intensity in the ratio between the total intensity and the local intensity. Denote ρ_j as the probability that the event j is an background event, and ρ_{ij} the probability that event j was triggered by event i. Then, we have

$$\rho_{ij} = \frac{\kappa(m_i)g(t_j - t_i)f(s_j - s_i; m_i)}{\lambda_\theta(x_j, y_j, t_j|\mathcal{H}_{t_j})}, \tag{1.11}$$

and

$$\rho_j = \frac{\mu(x_j, y_j)}{\lambda_\theta(x_j, y_j, t_j|\mathcal{H}_{t_j})} = 1 - \sum_{i<j} \rho_{ij}. \tag{1.12}$$

Thus, the correct estimation of θ should in turn help estimate whether an earthquake was triggered by the background rate, or if it is an aftershock of another

previous event in the catalog. When all the events in the catalog are far from the spatio-temporal border of the observed window W, we can consider the approximation $G(T-t_i) = F(S-s_i;m_i) \approx 1$, and with the results from Equations (1.11) and (1.12), we can substitute the indicator function for the unobserved u_j's in Equation (1.10) for their estimated probabilities as such

$$\begin{aligned} l_2(\theta|\mathcal{H}_t) = &\sum_{j=1}^{n} \rho_j \log\Big(\mu(x_j,y_j)\Big) + \sum_{i=1}^{n}\sum_{i<j} \rho_{ij} \log\Big(\kappa(m_i)g(t_j-t_i)f(s_j-s_i;m_i)\Big) - \\ &T\int\int_S \mu(x,y)dx\,dy + \sum_{i:t_i<t} \kappa(m_i). \end{aligned} \tag{1.13}$$

This form for the log-likelihood is easier to maximize given it's additive terms, but requires the calculated probabilities ρ_{ij}. The computational cost remains $\mathcal{O}(n^2)$,

Part II

Proposed Methodology

2 Parameter Estimation

TODO explicitar o pipeline do processo: apresentação de uma base de dados, uso do pacote ETAS, etc.

2.1 EM-Type Algorithm

For the estimation of the parameter vector θ, Veen & Schoenberg (2008) propose an EM-type approach, in which the probabilities ρ_j are updated in the E-step and the simplified complete log-likelihood (as shown in Equation (1.13)) is maximized in the M-step. Since the β parameter, who controls the distribution of the magnitudes, can be estimated analytically without interfering in the estimation of θ, it will be not included in the EM scheme.

Consider a catalog containing n earthquakes above a specified cutoff magnitude m_0, arranged by event times, and for each event $i \in \{1, 2, \ldots, n\}$, the catalog contains their longitudes and latitudes, magnitudes and time (in days) since the start of the monitoring T_0 (as presented in Table 1). Let $\Delta t_{ij} = (t_j - t_i), i < j$, and $\Delta m_i = (m_i - m_0)$.

To initiate the algorithm, an initial guess for θ must be given, as well as an initial guess for $\mu(x, y)$ in all the study region. In their work, Veen & Schoenberg (2008) separate S in arbitrary blocks and treat $\mu(x, y)$ as different constants in each block, but for this work, we divide S in a grid, in which every cell will be the discretization of the function rate evaluated in it's centroid. For the initial guess, we assume that the background rate function is constant in all the region, in a way that $\mu^{(0)}(x, y) = \dfrac{1}{N_{grid}}$, where N_{grid} is the number of cells in which S were divided.

In Jalilian (2019), the author cautions that when using the `etasfit()` function from the package `ETAS`, the algorithm used for the parameter estimation is very sensitive for the initial guess of $\theta^{(0)}$, to the point of warning the user when running the function. The sensitivity to the initial guess of the proposed EM-type algorithm will be further explored in section 3.1.

2.1.1 E-Step

The estimation of the probabilities that event i triggered event j is done using the expression given by Equation (1.12)

2.1.2 M-Step

The Q function to be maximized is

$$Q(\theta, \mu|\theta^{(k)}, \mu^{(k)}) = \underbrace{\sum_j \rho_j \log\left(\mu^{(k)}(x,y)\right) - T\int\int_S \mu^{(k)}(x,y)dx\,dy}_{Q_\mu} + \underbrace{\sum_j \sum_{i<j} \rho_{ij} \log\left(\kappa(m_i)g(\Delta t_{ij})f(s_j - s_i; m_i)\right) - \sum_i \kappa(m_i)}_{Q_\theta} \tag{2.1}$$

2.1.2.1 Analytical Gradient for Q

In most computational maximization routines based in Newton/quasi-Newton methods, providing the gradient of the objective function eliminates the need to estimate it by finite differences methods, improving significantly the computational time necessary for the convergence of the method.

The logarithmic part of the Q_θ function can be separated into independent blocks that depend only on some of the parameters, due to the parametrization of λ_θ, as such

$$l_{Q_\theta} = \log(\kappa(m_i)g(\Delta t_{ij})f(s_j - s_i; m_i)) = \underbrace{\log(\kappa(m_i))}_{A,\alpha} + \underbrace{\log(g(\Delta t_{ij}))}_{c,p} + \underbrace{\log(f(s_j - s_i; m_i))}_{D,\gamma,q},$$

where A and α are responsible for the productivity, c and p for the temporal aspects of the process and D, γ and q for the spatial distribution. This makes the derivatives in relation to each variable in θ to be dependent only of itself and the other variables in it's block, in a way that the parameters only interfere on the aspects they are supposed to control. If approximation methods for the gradient based in finite differences were used, the resulting hessian matrix would have crossed terms close to 0, but not really 0.

Let $r_{ij}^2 = \left[(x_j - x_i)^2 + (y_j - y_i)^2\right]$, $S_i = D\exp(\gamma \Delta m_i)$, $V_{ij} = \dfrac{r_{ij}^2}{S_i}$, $U_{ij} = 1 + V_{ij}$ and $W_{ij} = q\dfrac{V_{ij}}{U_{ij}} - 1$, for any events i and j such that $i < j$. With these expressions, the derivatives of l_{Q_θ} with relation to each variable are

$$\begin{aligned}
&\frac{\partial l_{Q_\theta}}{\partial A} = \frac{1}{A}, && \frac{\partial l_{Q_\theta}}{\partial \alpha} = \Delta m_i,\\
&\frac{\partial l_{Q_\theta}}{\partial c} = \frac{p-1}{c} - \frac{p}{\Delta t_{ij} + c}, && \frac{\partial l_{Q_\theta}}{\partial p} = \frac{1}{p-1} + \log\left(\frac{c}{\Delta t_{ij} + c}\right),\\
&\frac{\partial l_{Q_\theta}}{\partial D} = \frac{W_{ij}}{D}, && \frac{\partial l_{Q_\theta}}{\partial \gamma} = \Delta m_i W_{ij},\\
&\frac{\partial l_{Q_\theta}}{\partial q} = \frac{1}{q-1} - \log(U_{ij}). &&
\end{aligned} \tag{2.2}$$

The compensator term for Q_θ, given by $\sum_i \kappa(m_i)$, depends only of A and α, and has derivatives given by

$$\frac{\partial}{\partial A}\sum_i \kappa(m_i) = \sum_i \exp(\alpha \Delta m_i), \qquad \frac{\partial}{\partial \alpha}\sum_i \kappa(m_i) = \sum_i A\Delta m_i \exp(\alpha \Delta m_i). \tag{2.3}$$

When we combine the expressions in Equations (2.2) and (2.3), with the probabilities calculated in the E-Step ρ_{ij}, we have an complete expresion for the gradient of the Q_θ function, given by

$$\begin{aligned}
\frac{\partial Q_\theta}{\partial A} &= \frac{1}{A}\sum_{i<j}\rho_{ij} - \sum_i \exp(\alpha \Delta m_i), \\
\frac{\partial Q_\theta}{\partial \alpha} &= \sum_{i<j}\rho_{ij}\Delta m_i - A\sum_i \Delta m_i \exp(\alpha \Delta m_i), \\
\frac{\partial Q_\theta}{\partial c} &= \sum_{i<j}\rho_{ij}\left(\frac{p-1}{c} - \frac{p}{\Delta t_{ij}+c}\right), \\
\frac{\partial Q_\theta}{\partial p} &= \sum_{i<j}\rho_{ij}\left(\frac{1}{p-1} + \log\left(\frac{c}{\Delta t_{ij}+c}\right)\right), \\
\frac{\partial Q_\theta}{\partial D} &= \frac{1}{D}\sum_{i<j}\rho_{ij}W_{ij}, \\
\frac{\partial Q_\theta}{\partial \gamma} &= \sum_{i<j}\rho_{ij}\Delta m_i W_{ij}, \\
\frac{\partial Q_\theta}{\partial q} &= \sum_{i<j}\rho_{ij}\left(\frac{1}{q-1} - \log(U_{ij})\right).
\end{aligned} \tag{2.4}$$

2.1.2.2 Smoothing Step

The unrestricted maximizer for Q_μ is the sum of the pontual masses with weights $\frac{\rho_j}{T}$ in each of the cells in the study region, becoming degenerated and thus needing some form of smoothing. For that, we use the adaptative bandwidth gaussian kernel density estimator (KDE) for $\mu(x, y)$, given by

$$\hat{\mu}(x,y) = \frac{1}{T}\sum_j \rho_j \frac{1}{2\pi h_j^2}\exp\left\{-\frac{(x-x_j)^2+(y-y_j)^2}{2h_j^2}\right\}, \tag{2.5}$$

where $h_j = \max\left(d_j^{(n_p)}, d_{\min}\right)$ is the adaptative bandwidth size around the j-th event. It is given by the maximum between some designated minimum distance $d_{\min}$, and the distance from that event to it's n_p-th nearest neighbor. This form is proposed by Zhuang et al. (2002), and in it the authors find that the values for n_p and $d_{\min}$ are not sensitive and do not interfere significantly in the final estimate of $\mu(x, y)$, when their values remains in the ranges 15-100 for n_p and 0.01-0.1 for $d_{\min}$. We adopt the values $n_p = 20$ and $d_{\min} = 0.02$ as suggested by the authors.

Each kernel integrates 1 in $\mathbb{R}^2$, so $T \int\int_S \hat{\mu}(x,y)dx\,dy$ is equal to the sum of ρ_j, except the mass that escapes the study region S. Because of this, it is important to assure that the border capture most events and few occur outside S. Another important consideration to be done is the resolution of the grid, because $\hat{\mu}(x,y)$ for some event occurring in (x^*, y^*) is evaluated as the value in the centroid of it's nearest cell. With that in mind, a finer grid improves the pontual evaluation of the background rate, but is prone to under/overestimate $\mu(x,y)$, while a coarser grid captures better the behavior of the mean of $\mu(x,y)$, but fails to capture the unique seismological aspects of the study region.

Given some fixed $\mu(x,y)$, the M-step in θ is a real M-step in the sense that if $Q\left(\theta^{(k+1)}, \mu^{(k)}|\theta^{(k)}, \mu^{(k)}\right) \geqslant Q\left(\theta^{(k)}, \mu^{(k)}|\theta^{(k)}, \mu^{(k)}\right)$, then the observed log-likelihood does not decrease. However, this smoothing step does not maximize Q_μ, so the complete scheme is an EM-algorithm with a smoothing step along the lines of the EMS algorithm (Silverman et al. (1990)), with no guarantee for joint monotonicity.

Algorithm 1 – EMS Algorithm for the Estimation of ETAS Model

Data: Catalog with n events $\{(x_i, y_i, t_i, m_i)\}_{i=1}^{n}$, cutoff magnitude m_0, final time T, grid S with N_{grid} cells, tolerances $\epsilon_Q, \epsilon_\mu, \epsilon_\theta$, hyperparameters $n_p = 20$ and $d_{min} = 0.02$.

Result: Estimates of $\hat{\theta} = (A, \alpha, c, p, D, \gamma, q)$ and $\hat{\mu}(x, y)$.

Estimate β analytically;

Initialize $\theta^{(0)}$ with some viable guess;

for *each cell in* S **do**

  $\mu^{(0)}(x, y) \leftarrow \dfrac{1}{N_{grid}}$;

end

$k \leftarrow 0$;

Convergence $\leftarrow$ False;

while $\neg$*Convergence* **do**

  `// E Step`

  for $j \leftarrow 1$ **to** n **do**

    Evaluate $\lambda_{\theta^{(k)}}(x_j, y_j, t_j | \mathcal{H}_{t_j})$ using $\theta^{(k)}$ e $\mu^{(k)}$;

    for $i \leftarrow 1$ **to** $j - 1$ **do**

      $\rho_{ij}^{(k)} \leftarrow \dfrac{\kappa(m_i) g(t_j - t_i) f(s_j - s_i; m_i)}{\lambda_{\theta^{(k)}}(x_j, y_j, t_j | \mathcal{H}_{t_j})}$;

    end

    $\rho_j^{(k)} \leftarrow 1 - \sum_{i<j} \rho_{ij}^{(k)}$;

  end

  `// M Step (with Smoothing)`

  $\theta^{(k+1)} \leftarrow \arg\max_{\theta} Q_\theta(\theta | \theta^{(k)}, \mu^{(k)})$; `// using quasi-Newton method and` ∇Q_θ

  for *each cell with centroid* (x, y) *in grid* S **do**

    for $j \leftarrow 1$ **to** n **do**

      $h_j \leftarrow \max(d_j^{(n_p)}, d_{min})$;

    end

    $\mu^{(k+1)}(x, y) \leftarrow \dfrac{1}{T} \sum_{j=1}^{n} \rho_j^{(k)} \dfrac{1}{2\pi h_j^2} \exp\left\{ -\dfrac{(x - x_j)^2 + (y - y_j)^2}{2h_j^2} \right\}$;

  end

  if $|Q(\theta^{(k+1)}, \mu^{(k+1)}) - Q(\theta^{(k)}, \mu^{(k)})| < \epsilon_Q$ **and** $|\mu^{(k+1)}(x, y) - \mu^{(k)}(x, y)| < \epsilon_\mu$ **and** $|\theta^{(k+1)} - \theta^{(k)}| < \epsilon_\theta$ **then**

    Convergence $\leftarrow$ True;

  else

    $k \leftarrow k + 1$;

  end

end

return $\hat{\theta} \leftarrow \theta^{(k+1)}$, $\hat{\mu}(x, y) \leftarrow \mu^{(k+1)}(x, y)$;

2.2 Computational Implementation

The proposed method was implemented in the R programming language, version 4.6.1 (R Core Team (2026))

2.3 Goodness of fit tests

Theorem 3 (Random time change)**.** *Let* $0 < t_1 < t_2 < \ldots < t_n$ *be the realization of a Point Process with conditional intensity function* $\lambda(t|\mathcal{H}_t)$*, where* $\lambda(t|\mathcal{H}_t) > 0$ *for all* $t \in [0, T]$*. Then, the transformed times:*

$$\{\Lambda(t_1), \Lambda(t_2), \ldots, \Lambda(t_n)\} = \{\tau_1, \tau_2, \ldots, \tau_n\},$$

are the points of a unidimensional unit rate Poisson Process, if $\Lambda(t|\mathcal{H}_t)$ *is almost surely finite for all* $t \in [0, T]$*.*

This theorem holds for spatio-temporal point processes, and it is used for goodness of fit tests.

3 Simulation Studies

3.1 Sensitivity to Initial Guess of $\theta^{(0)}$

4 Case Studies

4.1 Iran Catalog

4.2 Peru Catalog

5 Final Considerations

Precisa de ajuda?

Consulte os manuais abnTEX2 (ARAUJO, 2015a; ARAUJO, 2015c; ARAUJO, 2015d) e da classe memoir (WILSON; MADSEN, 2010).

Envie um e-mail para `gfabinhomat@gmail.com` (Fábio Rodrigues Silva), que é o responsável pela manutenção desta versão de adaptação do modelo canônico abnTEX2.

Consulte a FAQ com perguntas frequentes e comuns no portal do abnTEX2: https://github.com/abntex/abntex2/wiki/FAQ.

Inscreva-se no grupo de usuários LATEX: http://groups.google.com/group/latex-br, tire suas dúvidas e ajude outros usuários.

Participe também do grupo de desenvolvedores do abnTEX2: http://groups.google.com/group/abntex2 e faça sua contribuição à ferramenta.

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Appendix

APPENDIX A – Cumulative distributions of $g(t)$ and $f(x, y)$

Considering the formulation of the probability density functions $g(t)$ and $f(x, y)$, in section 1.2, we can calculate their cumulative distribution function as such

$$\begin{aligned}
G(T) = \int_0^T g(t)dt &= \int_0^T (p-1)c^{p-1}(t+c)^{-p}dt \\
&= (p-1)c^{p-1}\int_0^T (t+c)^{-p}dt \\
&= (p-1)c^{p-1}\left[\frac{(t+c)^{1-p}}{1-p}\Big|_0^T\right] \\
&= (p-1)c^{p-1}\left[\frac{(T+c)^{1-p}}{1-p} + \frac{c^{1-p}}{p-1}\right] \\
&= 1 - \frac{(p-1)c^{p-1}}{(p-1)}(T+c)^{1-p} \\
&= 1 - \frac{c^{p-1}}{(T+c)^{p-1}} \\
&= 1 - \left(\frac{c}{T+c}\right)^{p-1}.
\end{aligned}$$

With this result, we can see that this is indeed a cumulative probability function, as $\lim_{T\to 0} G(T) = 0$ and $\lim_{T\to\infty} G(T) = 1$. For $f(x, y)$, consider the spherical coordinates $x = r\cos(\theta)$ and $y = r\text{sen}(\theta)$ and $D[\gamma(m - m_0)] = Q$. With those, we have

$$\begin{aligned}
F(R) = \int_0^{2\pi}\int_0^R f(r)dr\,d\theta &= \int_0^{2\pi}\int_0^R \frac{(q-1)}{\pi Q}\left(1+\frac{r^2}{Q}\right)^{-q} r\,dr\,d\theta \\
&= \frac{(q-1)}{\pi Q}\left(\int_0^{2\pi} d\theta\right)\left(\int_0^R r\left(1+\frac{r^2}{Q}\right)dr\right) \\
&= \frac{2(q-1)}{Q}\int_0^R r\left(1+\frac{r^2}{Q}\right)dr.
\end{aligned}$$

Let $u = 1 + \frac{r^2}{Q}$, so that $du = \frac{2r}{Q}dr$ and $r\,dr = \frac{Q}{2}du$. Using substitution, we have

$$
\begin{aligned}
F(R) &= \frac{2(q-1)}{Q} \int_1^{1+\frac{R^2}{Q}} u^{-q} \left(\frac{Q}{2}\right) du \\
&= (q-1) \int_1^{1+\frac{R^2}{Q}} u^{-q} du \\
&= (q-1) \left(\frac{u^{1-q}}{1-q} \Big|_1^{1+\frac{R^2}{Q}} \right) \\
&= -u^{1-q} \Big|_1^{1+\frac{R^2}{Q}} \\
&= 1 - \left(1 + \frac{R^2}{Q}\right)^{1-q},
\end{aligned}
$$

that is a valid cumulative probability function, as $\lim_{R^2 \to 0} = 0$ and $\lim_{R^2 \to \infty} = 1$.

APPENDIX B – Quisque libero justo

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Annex

ANNEX A – Morbi ultrices rutrum lorem.

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ANNEX B – Cras non urna sed feugiat cum sociis natoque penatibus et magnis dis parturient montes nascetur ridiculus mus

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ANNEX C – Fusce facilisis lacinia dui

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